

SCHOOL OF COMPUTER SCIENCE AND IT

YBN UNIVERSITY RANCHI

JAVA PROGRAMMING

ASSIGNMENT 08

Problem 1: Matrix Addition

Description:

Write a Java program that performs the addition of two 2x2 matrices. The program should prompt the user to input elements for two 2x2 matrices, then add the corresponding elements of the matrices and store the result in a third matrix. Finally, the program should print the resulting matrix.

Concepts Covered:

- Creation and initialization of a 2D array
- Accessing and modifying elements in a 2D array
- Looping through a 2D array

Hint:

- Use nested `for` loops to iterate through the rows and columns of the matrices and perform element-wise addition.

Problem 2: Transpose of a Matrix

Description:

Write a Java program that takes a 3x3 matrix as input from the user and computes its transpose. The transpose of a matrix is obtained by swapping the rows and columns of the original matrix.

Concepts Covered:

- Creation and manipulation of a 2D array
- Accessing elements using different indices
- Looping through a 2D array

Hint:

- Use nested `for` loops to iterate through the matrix and swap the elements at indices `[i][j]` with `[j][i]`.

Problem 3: Find the Sum of Each Row and Column in a Matrix

Description:

Write a Java program that prompts the user to input a 3x3 matrix and then calculates and prints the sum of each row and each column of the matrix.

Concepts Covered:

- Looping through rows and columns in a 2D array
- Accessing and manipulating elements in a 2D array

Hint:

- Use nested `for` loops to iterate through the rows and columns separately, maintaining separate sums for rows and columns.